Run report: N83 precomputed-table rho control, run 2 (38878565324)

Workflow `.github/workflows/n83-precomputed-rho-control.yml`, commit `44f63a66`, GitHub Actions run 38878565324, started 2026-10-16T09:09 UTC. Two jobs (k3300, k600), each on one hosted `ubuntu-24.04` runner (4 vCPU, shared, L0). Arms per job: IC (frozen producer, one process, 16 targets), empty-table fold rho (16 processes), precomputed rho (a) 650 s, (a′) = IC wall, (b) = IC max RSS. Targets: the 16 public points of `gha-38597929540-1`. Requested 16 targets per arm per job; completed 16 per arm per job, no failures or caps. Raw records: `records/run-38878565324/`. Protocol: `PROTOCOL.md`. Observations only; the verdict is in `REPORT.pdf`.

run 2 (38878565324) — IC K600 job (Intel(R) Xeon(R) 6973P-C)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: **`true`** (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	563.884	[165.222, 1277.173]	1966.983	[582.617, 3574.033]
rho, empty table (fold, Tier 1 worker)	1340.162	[927.073, 2401.124]	1620.925	[1182.901, 2092.605]
rho + table, (a) 650 s, seed …601	2.337	[1.406, 2.83]	2.406	[1.808, 3.057]
rho + table, (a′) = IC wall, seed …605	5.172	[3.381, 9.073]	5.91	[4.396, 7.622]
rho + table, (b) = IC max RSS	1.189	[0.887, 1.773]	1.729	[1.074, 2.548]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	296.64	[64.68, 657.79]	817.56	[232.1, 1619.86]	16/16
ic / bl_a2_wall_equal	81.52	[46.35, 236.59]	332.84	[89.97, 708.38]	16/16
ic / bl_b_memory_matched	395.53	[177.93, 1134.45]	1137.58	[309.35, 2697.39]	16/16
rho_plain_fold / bl_a_time_matched	786.74	[387.16, 1112.35]	673.72	[459.69, 932.67]	16/16
rho_plain_fold / ic	3.62	[0.93, 4.48]	0.82	[0.4, 2.82]	11/16

precomputation	wall s	CPU s	memory	notes
IC	773.7	804.03	5.75 GB max RSS	rank threads 1
rho + table, (a) 650 s, seed …601	650.8	651	0.438 GB table	27354500 entries, 5.37 M steps/s, stop: wall_seconds
rho + table, (a′) = IC wall, seed …605	772.7	773	0.512 GB table	31981512 entries, 5.29 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	1904.4	7558	5.67 GB table	354379300 entries, 11.97 M steps/s, stop: table_bytes

run 2 (38878565324) — IC K3300 job (AMD EPYC 7763 64-Core Processor)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: **`true`** (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	32.784	[24.2, 67.529]	49.256	[31.481, 70.282]
rho, empty table (fold, Tier 1 worker)	1967.327	[1326.671, 3570.145]	2375.319	[1739.184, 3058.48]
rho + table, (a) 650 s, seed …601	3.396	[2.032, 4.739]	4.205	[2.749, 6.052]
rho + table, (a′) = IC wall, seed …605	8.113	[6.261, 12.137]	8.913	[6.826, 11.042]
rho + table, (b) = IC max RSS	1.167	[0.497, 1.598]	1.183	[0.836, 1.568]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	11.33	[4.08, 18.8]	11.71	[6.7, 21.47]	15/16
ic / bl_a2_wall_equal	3.74	[2.65, 11.05]	5.53	[3.31, 9.03]	14/16
ic / bl_b_memory_matched	40.61	[24.27, 71.72]	41.63	[24.71, 71.95]	15/16

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
rho_plain_fold / bl_a_time_matched	803.85	[401.15, 1115.98]	564.83	[330.72, 918.76]	16/16
rho_plain_fold / ic	70.9	[19.17, 92.37]	48.22	[26.71, 87.87]	16/16

precomputation	wall s	CPU s	memory	notes
IC	681.6	825.02	10.79 GB max RSS	rank threads 4
rho + table, (a) 650 s, seed …601	650.6	651	0.304 GB table	19008681 entries, 3.73 M steps/s, stop: wall_seconds
rho + table, (a′) = IC wall, seed …605	680.2	680	0.317 GB table	19825177 entries, 3.72 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	5293	21036	10.519 GB table	657416242 entries, 8.07 M steps/s, stop: table_bytes

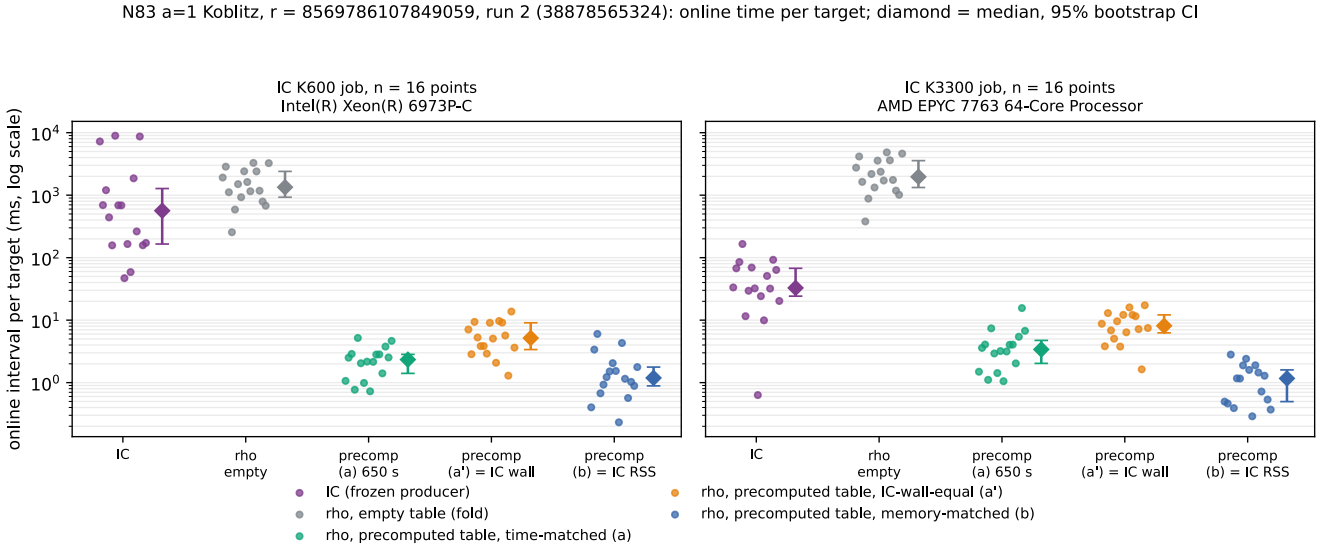


Figure 1: run 2 (38878565324): online interval per target; diamond = median with 95% bootstrap interval.